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Developing a personalized meal recommendation system for Chinese elderly population: a knowledge-driven and community-based approach

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Abstract

Background: Knowledge graph-based food recommendations are critical for the nutritional support of older adults. However, the integration of disease-based nutritional principles and preference-related requirements needs to be optimized in current recommendation processes.

Objective: This study aims to design a knowledge graph-based personalized meal recommendation system for community-dwelling elders, and to conduct the preliminary effectiveness testing.

Methods: ElCombo, a personalized meal recommendation system, was developed driven by user profiles and food knowledge graph. User profiles were established from a survey of 96 community-dwelling elders. Food knowledge graphs were supported by data from websites of Chinese cuisine recipes and eating history, consisting of five entity classes (dishes, ingredients, category of ingredients, nutrients, and diseases), corresponding attributes, and relations between entities. A personalized meal recommendation algorithm was then developed to generate packaged meals as outputs, considering disease-related nutritional constraints and personal dietary preferences.

Results: Our recommendation system, ElCombo, was evaluated by comparing the dietary diversity and diet quality of its recommended meals with those of the autonomous choices of community-dwelling elders. Simulation experiments based on retrospective data of 96 community-dwelling elders revealed that recommended meals had significantly higher diet quality and dietary diversity ($P < 0.001$). Two representative cases involving community-dwelling elders with and without eating history demonstrated the recommendation system's potential to fulfill complex nutritional requirements associated with multiple morbidities.

Conclusions: ElCombo proved superior performance in simulations compared to autonomous choices, implying the potential for improving dietary practices for community-dwelling elders. Future studies are needed to optimize its real-world application and refine data handling abilities.

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Abstract

Background: China's aging population is facing serious health challenges, including malnutrition and multiple chronic conditions. There's a critical need for tailored food recommendation systems. Knowledge graph-based food recommendations offer considerable promise in delivering personalized nutritional support. However, the integration of disease-based nutritional principles and preference-related requirements needs to be optimized in current recommendation processes.

Objective: This study aims to develop a knowledge graph-based personalized meal recommendation system for community-dwelling elders, and to conduct the preliminary effectiveness testing.

Methods: ElCombo, a personalized meal recommendation system, was developed driven by user profiles and food knowledge graph. User profiles were established from a survey of 96 community-dwelling elders. Food knowledge graphs were supported by data from websites of Chinese cuisine recipes and eating history, consisting of five entity classes: dishes, ingredients, category of ingredients, nutrients, and diseases, along with their attributes and interrelations. A personalized meal recommendation algorithm was then developed to synthesize this information to generate packaged meals as outputs, considering disease-related nutritional constraints and personal dietary preferences. Furthermore, a validation study employing a real-world dataset collected from 96 community-dwelling elders, was conducted to assess ElCombo's effectiveness in modifying their dietary habits over a one-month intervention, using simulated data for impact analysis.

Results: Our recommendation system, ElCombo, was evaluated by comparing the dietary diversity and diet quality of its recommended meals with those of the autonomous choices of 96 eligible community-dwelling elders. Participants were grouped based on whether they had a recorded eating history, with 34 having and 62 lacking such data. Simulation experiments based on retrospective data over a 30-day evaluation revealed that ElCombo's meal recommendations consistently had significantly higher diet quality and dietary diversity compared to the elders' own selections ($P < .001$). Additionally, case studies of two elders, one with and one without prior eating records, showcased ElCombo's ability to meet to fulfill complex nutritional requirements associated with multiple morbidities, personalized to each individual's health profile and dietary requirements.

Conclusions: ElCombo has shown enhanced potential for improving dietary quality and diversity among community-dwelling elders in simulation tests. The evaluation metrics suggest that the food choices supported by the personalized meal recommendation system surpass autonomous selections. Future research will focus on validating and refining ElCombo's performance in real-world settings, emphasizing the robust management of complex health data. The system's scalability and adaptability pinpoint its potential for making a meaningful impact on the nutritional health of the

aging population.

Key words: knowledge graph, personalized food recommendation, elder nutrition, community



Introduction

With the global trend of an aging population [1], China leads the largest elderly population proportion in the world, expecting 28% of its population to be over 60 by 2040 [2]. An estimated 25.3% of China's elderly are diagnosed with multiple chronic conditions (MCCs) [3]. Nutrition plays a critical role in healthy aging, contributes significantly to mitigating age-associated comorbidities [4]. However, according to the Chinese Longitudinal Healthy Longevity Survey (CLHLS) results [5], 35% elderly population (>65 years old) are at risk of malnutrition [6]. This is largely attributed to inappropriate eating behaviors, particularly a lack of food diversity and poor diet quality [7], which can further lead to nutritional deficiencies and dietary disequilibrium [8, 9], thereby negatively exacerbating MCCs and affecting overall wellbeing [6, 10].

Currently, approximately 45% of the Chinese elderly population (>65 years) are community-dwellers (i.e. outside nursing homes or hospitals) who live alone or with only one spouse in home environments [11, 12]. They face the challenges like reduced physical function and altered sensory perceptions, which impact their dietary habits and nutritional intake [1, 13, 14]. These issues, along with cultural and literacy barriers [14-16], collectively reduce their adherence to a diverse, high-quality diet [17]. Several strides have been made in promoting healthy eating among community-dwelling elders, primarily through health education and providing uniform nutrient supplement or food services [18, 19]. However, these interventions often stick to the dietary guidelines for general populations. Their focuses on adequate consumption of certain food groups and nutrients might not be applicable to elderly individual needs and preferences [16, 20]. While some interventions offer flexibility by involving clinicians or nutritionists for personalized dietary recommendations or monitoring [21], the restricted professional and information resources are insufficient not sustainable in addressing the community-dwelling elders' complex nutritional needs associated with MCCs [20, 22].

Elderly eating behaviors are influenced by multiple factors, including physical and physiological factors, age-related issues, and social determinants [1, 23]. Ideal diets for the elderly should, therefore, be personalized, considering individual health, activity levels, and preferences, and guided by expert knowledge [23] [16]. Simultaneously, addressing these needs effectively is a central objective of personalized food recommendation systems.

Table 1. A summary of food recommendation system types, applied algorithm and application scenarios

Food recommendation system type	Application scenarios	Recommendation approaches	Reflections
Type 1: User Preferences	online food ordering	Content-based,	Tailored to user tastes

Focused [24-26]	platforms, recipe websites.	Knowledge-based, Collaborative filtering, Hybrid approach	for high satisfaction, it may neglect nutrition and variety beyond past preferences.
Type 2: Nutritional Needs Focused [27-29]	Specialized meal plans for patients with specific health conditions, diet therapy.	Constraint-based, Goal-oriented recipe, Knowledge-based	It addresses health and dietary needs but may not match user tastes and depends on precise health data.
Type 3: Balancing Preferences and Needs[30, 31]	home meal planning, health-conscious food apps.	Constraint-based, Content-based, Knowledge-based	Balances enjoyment with health, yet managing this balance poses a complexity.

Table 1 categorizes the main types of food recommendation systems by focus area, applications, and approaches, highlighting their respective strengths and limitations. Type 1, focused on user preferences, recommends dishes similar to those previously enjoyed by the user, using collaborative filtering and content-based algorithms for item rating and user profile similarity computations. It's most applicable in personal dining and online food platforms, where individual tastes drive the selection process. Type 2 emphasizes nutritional needs, often using constraint-based approaches to recommend health-conscious meals, catering to users with specific dietary requirements or health conditions. Type 3 highlights a balance between user preferences and nutritional needs. This is often applied in scenarios like home meal planning or health-focused food apps. Despite a diverse focus, personalized food recommendation systems generally share similar algorithms, including collaborative filtering, content-based, and knowledge-based Systems. Collaborative filtering algorithms utilize user ratings to match and recommend items liked by similar users. They excel in user preference prediction but faces the cold start problem with small sample sizes. Content-based systems, which focus on item features and user history, achieve personalization but may limit diversity by predominantly suggesting previously preferred items. Knowledge-based systems recommend based on explicit user requirements and deep domain knowledge. They are effective in scenarios with sparse user data and particularly offer high customization in well-defined use preference settings or specialized domains like healthcare. In practice, a hybrid approach is more often applied to mitigate individual limitations. Specifically, in community-dwelling elderly food recommendations targeting a relatively small number of users, the integration of nutritional and health considerations is paramount. This context indicates that knowledge-based recommendations as

suitable choices, with domain expertise ensuring dietary appropriateness. A complementary role can be played by content-based systems, which align recommendations with individual taste profiles, together delivering nutritionally sound and appealing food choices.

A Food Knowledge Graph (food-KG) can be viewed as a graph-structured data base, aiming to organize food-related entities (data nodes), attributes, and the connected edges (relations) between them in a network of interconnected entities [32, 33]. They serve as integral components of knowledge-based food recommendation systems, organizing interconnected food-related entities to facilitate contextually appropriate dietary suggestions [33]. Personalized food recommendation systems based on a food KG aim to synthesize comprehensive personal information to customize dietary recommendations, which has the strength of providing sharable and uniform representations of food and nutrition knowledge to address the problem of information silos across multiple sources [33, 34]. While these systems have been implemented successfully across various sectors of the food industry, such as personalized diet therapy planning for patients with specific diagnoses and food and restaurants recommendations to the general population [32, 34-40], their application often lacks specificity for community-dwelling elderly, a demographic with unique dietary needs due to multimorbidity and personal preferences. Currently, there is a scarcity of Food KGs and corresponding KG-based food recommendation systems specifically designed to meet the unique dietary needs and health considerations of this demographic. The absence of targeted data, such as actual meal consumption data and primary data of these elders' health status and dietary preference, leads to this significant gap. This challenge highlights the necessity of developing a customized Food KG-based focused on the community-dwelling elders, incorporating primary data that captures the nuances of their health status and dietary choices. Consequently, a personalized recommendation system built on this Food KG should effectively manage the complex the interplay between dietary restrictions due to elderly MCCs and personal dietary preferences, which is a key aspect for community-based daily nutrition support targeting the elderly [31].

Finally, considering the alignment of current food recommendation systems with their target users, the prevalent method of offering a wide variety of dish options may not be suitable for the elderly. Community-dwelling elders generally have lower nutrition literacy compared to younger users who are the primary audience for most online food or restaurant recommendations. Additionally, they do not have nutrition knowledge support like hospitalized patients receive from clinical nutritionists. This lack of nutrition knowledge makes it challenging for them to select from abstract food recommendations, understand appropriate portion sizes, and create nutritionally balanced meals [34, 39, 41, 42]. Addressing the unique dietary needs of community-dwelling elders requires a nuanced

approach that considers their lower nutrition literacy and lack of nutrition knowledge support compared to other demographics. Digital interactions such as virtual assistants offer potential for user-friendly support, yet their integration into food recommendation systems for the elderly is still emerging. Thus, a deeper study of food-KG-based meal recommendations is warranted to meet the personalized nutritional needs of the elderly in their daily diets. Accordingly, we identified three key questions: 1) how to address the complex nutritional requirement associated with multimorbidity; 2) how to integrate personal dietary preferences into rigorous nutritional principles; 3) how to ensure user-friendliness for elderly users.

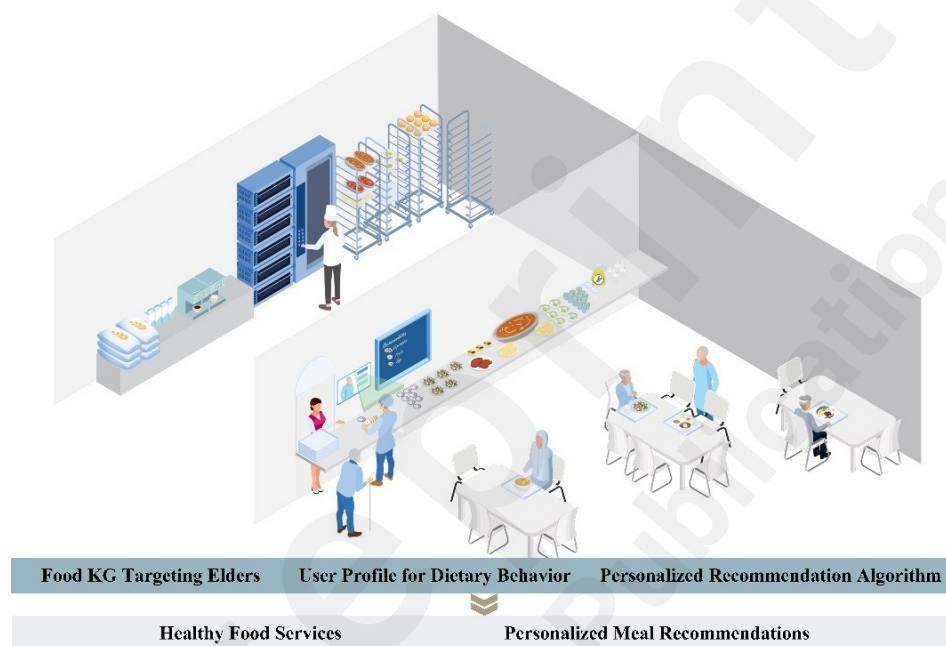


Figure 1. A scenario landscape of ElCombo application in personalized meal recommendation community-dwelling elderly care

In response to these considerations, the primary objective of this study is to propose “ElCombo”, short for “Elderly Combo”, a personalized meal recommendation system specifically designed for community-dwelling elders. **Figure 1** illustrates the conceptualized application scenario of the ElCombo system among community-dwelling elders, depicting the workflow from Food Knowledge Graph construction to the delivery of personalized meal recommendations. ElCombo aims to offer eating choices in the form of meal combos, which are combinations of several dishes. It considers both disease-related nutritional requirements and personal dietary preferences of community-dwelling elders, while also providing a user-friendly delivery mode. The secondary aim is to examine the effectiveness of ElCombo in improving the diet diversity and quality of community-dwelling elders, using primary data collected in this study. Our overarching aim is to assist these elders, especially those in home environments with limited access to professional clinical support, in making informed decisions faced with overwhelming information presented by advanced

technologies, consequently contributing to promoting healthy aging.

Methods

Guided by a user-centered design concept, ElCombo was designed based on the needs and knowledge acquired from real-world data, added with the recommendation outputs in the format of packaged meals, i.e., combinations of several dishes, to reduce the mental loads of community-dwelling elderly users.

ElCombo System Architecture

Figure 2 presents the architecture of personalized food recommendation system targeting community-dwelling elders. As shown in the diagram, ElCombo is consisted of 3 key modules: (1) Food knowledge graph (food-KG) targeting community-dwelling elders, which served as a knowledge base to collect and store information on candidate dishes, relevant ingredients, nutrients, and common geriatric diseases and disease-related constraints. Since the food-KG involved the experts' knowledge of food composition and food intake criteria informed by the diagnosis, it also supported to regulate the whole recommendation pipeline.

(2) The user profile dataset targeting elderly dietary behaviors, which captured personal needs related to disease-specific nutritional requirements and dietary preferences. It includes demographics, nutrition and health status, and dietary preference information of community-dwelling elderly individuals. This user profile dataset served as a key input for a nutritional recommendation approach. It leveraged data from survey results of community-dwelling elders, along with eating history behavior tracked by the IoT system deployed at the canteen (if applicable).

(3) The personalized meal recommendation algorithm, which generated tailored meal plans for community-dwelling elders. It combined nutritional expert knowledge and individual dietary needs, using similarity calculations and rule-based filters to create user-friendly packaged meals. The algorithm was driven by three factors: nutritional context for initial food filtering, short-term goals for daily meal generation based on user preferences and needs, and long-term goals adjusting recommendations according to eating history.

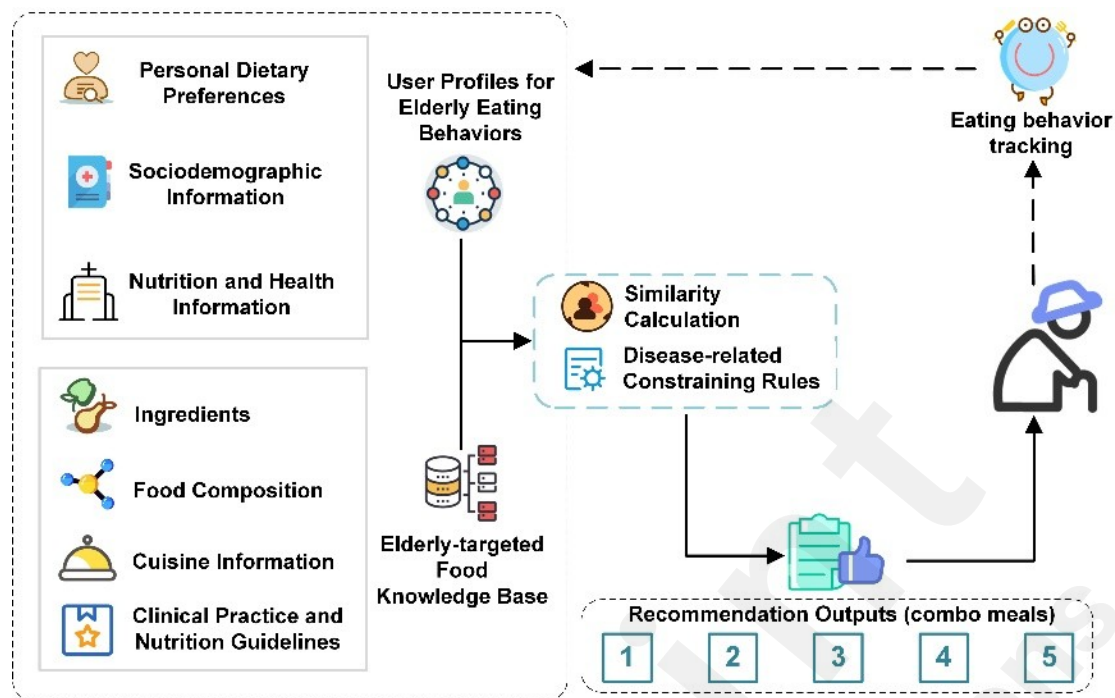


Figure 2. Overview of ElCombo's architecture for elder-specific food recommendation, including knowledge graph and algorithm modules

Initial data preparation

The initial steps necessary to prepare the data to be used in meal recommendation generation were aimed at identifying appropriate datasets to achieve two intermediate goals: 1) constructing food-KG, and 2) building user profiles related to elderly dietary behaviors.

Data sources

Six data sources were identified to contribute to the food-KG construction from three aspects. Firstly, food and recipe data were gathered from two sources: historical eating choices from local community-dwelling elder canteens, and various recipe websites. Secondly, food composition and nutrient information were sourced from a specific database Chinese Food Composition Table and the Dietary Nutrient Reference Intake for Chinese Residents [56, 67]. Lastly, common geriatric diseases and corresponding nutrition restrictions were collected from textbook and clinical guidelines related to the included diseases.

Data collection and preprocessing

To make use of the information from food and recipe websites, we first converted data from its original semi-structured HTML format on web pages into a more structured, tabular format suitable. This was done using the Scrapy framework in a Python 3.8 environment, where we extracted specific information from web pages, including dish names, ingredients, seasonings, cooking methods, and flavors. Once extracted, we conducted the data preprocessing by converting these details into a structured table format. The transition from web page representations of dishes to their tabular format

in data sheets is shown in **Figure 1** in **Appendix 1**, using ‘Stir-fried Spinach’ as an example.

To prepare for the real-word recommendation, we also included information on 42 dishes informed by the tracked eating history of community-dwelling elders. These dishes were already formatted into tables with details such as dish names, ingredients, and cooking methods. We manually removed 30 dishes that were receptive to the dishes obtained from websites, then finalized a list of 180 unique dishes for building our food Knowledge Graph (food-KG), as explained in **Figure 2** in **Multimedia Appendix 1**. In second phase of data preprocessing, we standardized the weights and ratios for the dishes, ingredients, and seasonings, as shown in **Table 1** in **Multimedia Appendix 1**. The nutrient content of each food item was calculated based on the standard amount of nutrients present in every 100 grams of the ingredients involved. This step linked all relevant data, including dishes, ingredients, ingredient categories, and nutrients, and prepared them for the next step in our recommendation process.

We took three main steps to gather and preprocess the data indicating the nutritional requirements of elderly individuals with geriatric diseases. The first step involved locating 30 common geriatric diseases across 12 human systems from a textbook of geriatric medicine [43]. Next, we identified statements within textbook and clinical guidelines that describe the relations between these diseases and various ingredients, ingredient categories, and nutrients. These guidelines cover a broad spectrum of common geriatric management topics such as cardiovascular and metabolic disease management, diabetes management, nutrition therapy for cancer patients, and dietary guidance for stroke patients [44-51].

The third step was the extraction of entities and relations from these statements. Regular expressions were applied considered the standardized and well-formatted statement expressions in the textbooks and clinical guidelines. We developed an exhaustive list of expressions of relations and corresponding variations, to systematically capture the relations and its triples in the statements. The outputs of this process include the main categories of relations, the triples containing the relations, examples of the statements extracted, and the sources from which these statements were derived (**Multimedia Appendix 1 Table 2**). This methodical approach helped organize the data on disease-nutrient interactions, facilitating the development of our nutritional recommendation framework for the elderly.

Food-KG Construction

Food-KG construction generally followed a top-down pipeline to align with authoritative guidelines and expert opinions (**Figure 3**). The food-KG construction process can be divided into four steps: 1) schema design, 2) data preparation, 3) knowledge extraction and reasoning, and 4) knowledge

storage and representation.

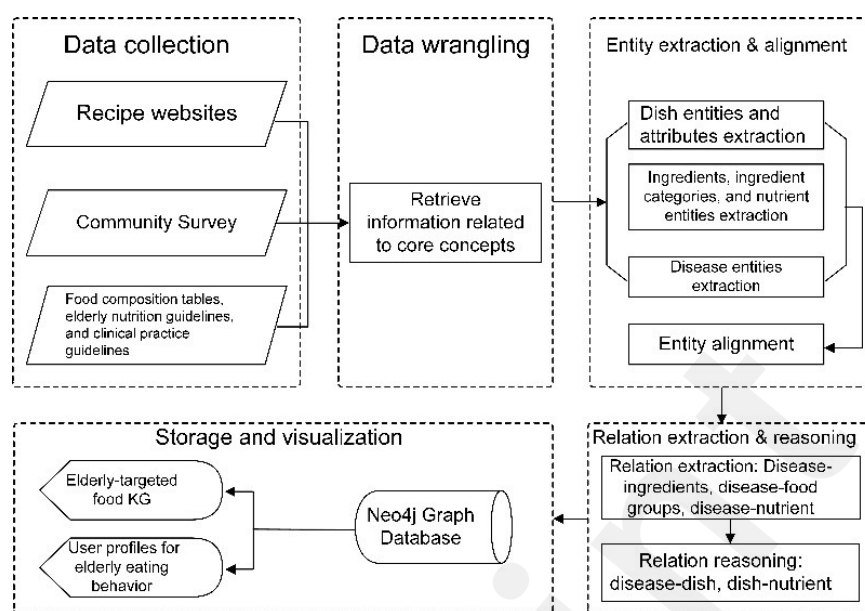


Figure 3. Food knowledge graph construction pipeline

Drawing on prior literature and expert opinions, we initially defined five central concepts to shape the framework of the Food-KG: dishes, ingredients, categories of ingredients, nutrients, and diseases. We also established their respective attributes and interrelationships. As depicted in **Multimedia Appendix 1 (Fig3, Table 3-4)**, the five pivotal concepts are interconnected. A 'dish' refers to any edible cuisine composed of various 'ingredients', which include raw materials and seasonings, thus establishing a 'consist_of' relation between 'dishes' and 'ingredients'. These 'ingredients' are grouped into different 'ingredient categories' in accordance with dietary guidelines, and relevant 'nutrients' are identified from food composition tables. 'Diseases' are incorporated as the final crucial concept, given their significant relevance to nutritional support for the elderly population.

Following the abovementioned five pivotal concepts, we obtained food, nutrition, and disease-related data from various sources and preprocessed it into a standard tabular format. We used the Scrapy framework to crawl "dishes" entities from the "elderly" section of recipe websites. At the same time, we extracted website data tagged with "cooking method" and "flavor" labels, serving as the attributes of "dish" entities. We included a "dish category" as an additional attribute to facilitate meal package recommendations. Furthermore, we extracted "main components" and "seasonings" labels attached to "dish" entities as "ingredients" and identified their interrelationships. We sourced "disease" entities from clinical practice guidelines and textbooks on common geriatric diseases. This process yielded a corpus of words, phrases, and sentences describing relationships between diseases and ingredients, food groups, and nutrients. During data preprocessing, we standardized the weight of each dish category and the proportions of seasonings and food contained. Specifically, we set a standard sodium resolution in each dish at $0.9 \pm 0.1\%$ to reduce the risk of hypertension in community-

dwelling elders.

We resolved potential name variations and ambiguities through knowledge fusion, implementing entity alignment and named entity disambiguation. We eliminated "dish" entity duplicates based on name similarity, included ingredients, and cooking methods. We set an ingredient similarity threshold at 85% and kept the most frequently occurring combinations as the standard for the corresponding "dish" entity. After removing desserts and snacks, we finalized 180 unique "dish" entities associated with unique sets of "ingredients" entities and attributes. We conducted the entity alignment of "ingredients" and "nutrient" entities according to the standard names listed on food composition tables. This resulted in 112 "ingredient" entities and 27 "nutrient" entities. We directly acquired "ingredient categories" and "disease" entity names from food nutrient intake instructions and textbooks. For these, we executed the entity extraction process by importing the standard names of 20 "ingredient categories" entities and 30 "disease" entities, bypassing the entity alignment.

Guided by the top-down design, the relationships between the entities of 'dishes', 'ingredients', 'ingredient categories', and 'nutrients' were established based on natural interactions. We identified three types of relationships between "diseases" and "nutrients", "ingredients", and "ingredient categories", namely discourage, restrict, and recommend. This categorization was synthesized from textbooks, clinical practice guidelines, and expert opinions. We employed rule-based relation extraction to identify disease-related triples, relying on a relation dictionary developed from our predefined relationship types, followed by manual checks to ensure accuracy.

Despite our extraction efforts, the relationships between “dishes” and “nutrients”, “dishes” and “ingredient categories”, and “diseases” and “dishes” remained undefined. We hypothesized that these relationships could be inferred from the existing entities and relations within our constructed food knowledge graph, denoted as $KG=\langle E,R,T \rangle$, and the relation path P . Here, E and T represent the set of entities, R represents the set of relations, and the edges in R link two nodes to form a triple $(h, r, t) \in T$. We sought to generate a new triple that didn't exist in the knowledge graph, denoted as $G=\{ (h, r, t) \mid h \in E, r \in R, t \in T, (h, r, t) \notin G \}$. This allowed us to infer new relationships and their properties that were missed during the initial extraction process. We performed this knowledge reasoning using a first-order logic reasoning method. For example, we used the formula: $\text{Contain}(\text{IngredientA}, \text{NutrientA}, M) \wedge \text{Consist_of}(\text{DishA}, \text{IngredientA}, P) \Rightarrow \text{Contain}(\text{DishA}, \text{NutrientA}, Q)$ to determine the relationship between a specific dish and nutrient entities (i.e., nutrient weights contained in a dish unit) by accumulating nutrients contained in different ingredients making up the dish according to proportions. The formula: $\text{Discourage}(\text{DiseaseA}, \text{IngredientA}) \wedge \text{Consist_of}(\text{DishA}, \text{IngredientA}) \Rightarrow \text{Discourage}(\text{DiseaseA}, \text{DishA})$ was used to determine relationships between specific diseases and

dishes. **Figure 4-a** shows the disease entity “gout” and its relations to relevant ingredients and ingredient categories.

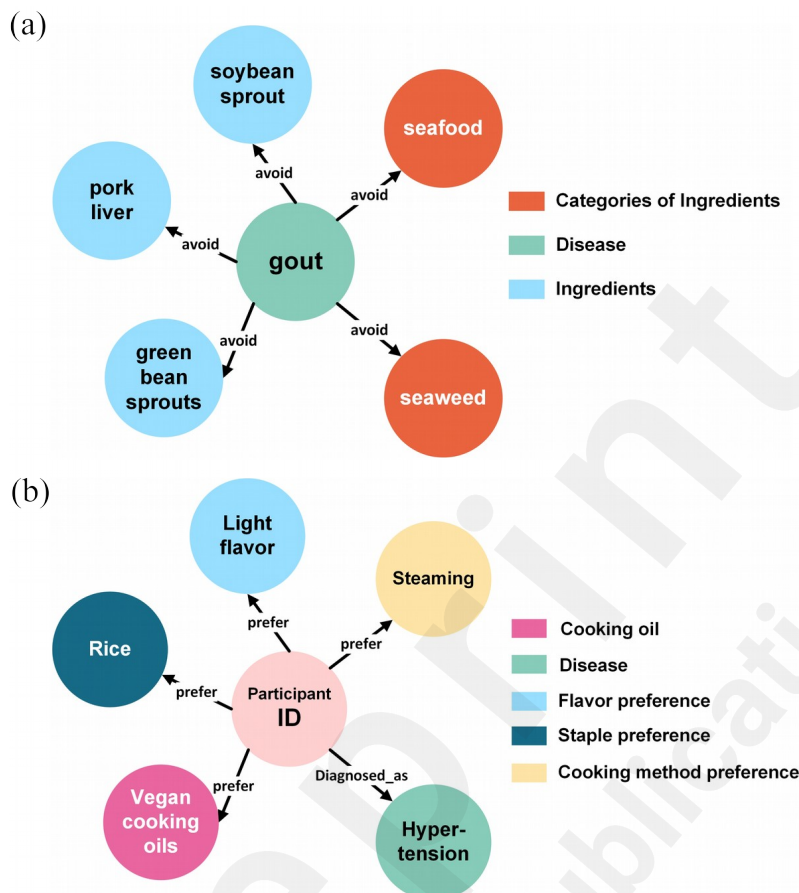


Figure 4. Visualizing disease-diet connections and individual health profiles in Food-KG. (a). Mapping 'gout' disease to relevant dietary ingredients and categories; (b). Elderly profile with hypertension, dietary preferences interconnections depicted.

The finalized food knowledge graph for the elderly (elderly food-KG) comprised 369 entities, 6 relationships, and 10,071 triples, stored in a Neo4j graph database. This data structure is a directed labeled graph consisting of nodes, edges, and labels. As illustrated in **Multimedia Appendix 1 (Figure 4)**, the entities of dishes (orange), ingredients (blue), and food category (red) function as nodes. An edge connects a pair of nodes and represents the interrelations, while the labels of the edge further specify the nature of these relations.

User profile Establishment

Paralleled with food-KG, we built user profiles for elderly eating behaviors to capture the individual characteristics of elderly target users, thus facilitating recommendation customization. Similarly, following a top-down workflow, we designed the architecture of user profiles including three dimensions- sociodemographic, disease and health conditions, and dietary preferences. We use survey data to get real-world status and actual needs. A total of 96 community-dwelling elders in

Hangzhou, China consented to participate in the survey. A total of 16 items were included in the questionnaire. Among them, the dietary preferences-related items were extracted from CHILS questions, and the sociodemographic and health condition items were created based on the medical records in the primary care center.

The user profiles were also stored into a Neo4j graph database to support entity and relation queries as a prerequisite for food recommendation. As shown in **Figure 4-b**, the pink node represents the entity of a community-dwelling elderly individual, the green nodes with the label of “hypertension” represent their diagnosis of disease and other nodes represented for the dietary preference from different perspectives. The edges between different nodes represent the property of interrelations.

Personalized Meal Recommendation Algorithms

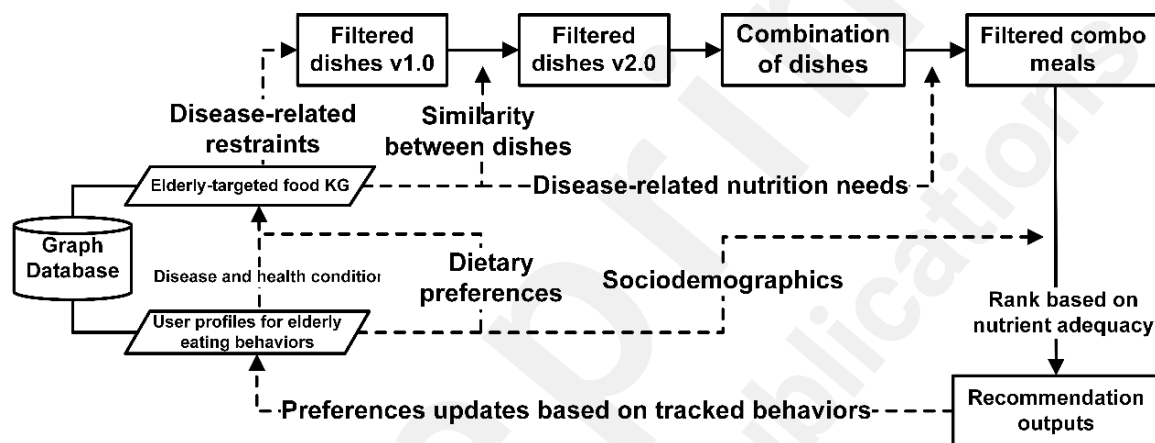


Figure 5. Dual-stage personalized meal algorithm: dish generation and combo meal recommendations

As shown in **Figure 5**, the personalized meal recommendation algorithm has two main components: candidate dish generation and combo meals recommendation.

The generation of candidate dishes can be divided into initial filtering and dish reranking. The initial filtering step aimed to eliminate dishes with ingredients unsuitable for elderly individuals due to disease-related dietary restrictions. It began with the application of disease-related restraints, informed by the information in food-KG and user profiles for elderly dietary behavior. Subsequent to this initial filtering, the algorithm further refined the selection of dishes by considering personal dietary preferences, this step considered two scenarios. (1) Hot boot for community-dwelling elders with tracked eating history: we employed a method to rank dish candidates based on their similarity to previously selected dishes. This ranking is derived by assessing the ingredient composition of dishes. Since ingredient content in dishes was quantified as continuous numerical data, Cosine similarity was adopted to assess the similarity between dishes. (2) Cold boot for community-dwelling elders without tracked eating history: we identify a reference elder with a closely matching user

profile. The assumption is that elders with similar profiles will have similar dietary preferences. The similarity between profiles is calculated based on their features. Because the user profile features were recoded as categorical variables, the Jaccard similarity measure was used for comparison. Also, as the recommendation duration get longer, the preferences and recommendation preference could be continuously updated based on tracked eating behaviors.

Combo Meal Recommendation focused on combining various dishes into meal combos that are nutritionally balanced and cater to the diverse dietary preferences and economic statuses of the elderly community. The process involved three key steps. (1) Initial Combo Meal Creation: Multiple dishes from different categories were involved to form initial combo meal candidates. This selection was guided by the goal of creating diverse combinations of dish categories at varying price levels, ensuring accessibility for community-dwelling elders of different socio-economic statuses. **Table 2** listed all possible combinations of different types of dishes. (2) Nutritional Constraints Assessment: The algorithm then evaluated the combo meals against the individual's nutritional needs. Meals that did not adhere to the requirements specified in nutrient guidelines, in terms of ingredient or ingredient category restrictions would be filtered out. (3) Final Combo Meal Ranking: The remaining meals were ranked based on their nutrient adequacy. This was measured by the number of nutrients that satisfied the criteria outlined in the Chinese elderly nutrition guidelines [55, 67].

Table 2 Dish combinations considered in in meal recommendation algorithm

Price	Dish counts	Dish Categories
High	4	1×animal-derived dish +1×vegan dish + 1×staples
High	4	2×omnivorous dishes +1×staples +1×soup
Intermediate high	3	1×animal-derived dish +1×vegan dish + 1×staples
Intermediate high	3	2×omnivorous dishes +1×staples
Intermediate	2	1×animal-derived/ omnivorous dish + 1× complete
Low	2	1×vegan dish + 1×complete dish

The detailed processes and calculations for candidate dish filtering and combo meal generation are provided as pseudocode in **Algorithms 1 and 2**, located in the **Multimedia Appendix 2**. The corresponding formulas for each calculation step are also included.

Validation study

In this study, we aim to evaluate the impact of the ElCombo meal recommendation system on the dietary behaviors of community-dwelling elders using a one-arm before-after design. To ensure safety and minimize risks, we will conduct a preliminary simulation test with primary data from elderly participants. The effectiveness of the intervention will be assessed based on changes in diet quality and diversity, measured using the China Elderly Dietary Guidelines Index (CDGI-E) [44, 52],

and the Diet Diversity Score (DDS) [53, 54].

Study setting and participants

To examine our hypothesis in validation study, we conducted a primary data collection among community-dwelling elders in Hangzhou City, Zhejiang Province, China, from July 15 to July 21, 2021. All participating community-dwelling elders and research team members were consented by the Institute of Medical Information, Chinese Academy of Medical Sciences and Peking Union Medical College (IMI, CAMS & PUMC). See details below in 'Ethical Considerations'.

We view this validation study as a one-arm before-after study. The power calculation is based on the primary outcome measure- diet diversity score (DDS). The sample size was calculated according to

the repeated-measures formula[10, 55, 56]: $N = \frac{2(Z_{1-\alpha/2} + Z_{1-\beta})^2 \sigma^2 (1-\rho)}{\delta^2}$, taking into account a Type

I error rate of 5% (two-sided), a power level of 80%, a smallest detected meaningful difference (δ) of 2.0 based on expert opinion), a correlation coefficient (ρ) among measures of 0.5, and an equal standard deviation (σ) of 6.0 were selected according to relevant studies [55, 57]. It was estimated that the desired number of included participants at baseline should be set at a minimum of 71 for each arm. Consequently, a total of 89 community-dwelling elderly individuals were expected to recruit at baseline considering an attrition rate of 20% [58-60].

Participants were selected using a convenience sampling method from community health service centers in Hangzhou. Elderly individuals were eligible if they were: (1) over 60, (2) able to communicate independently, (3) able to make independent food choices; (4) able to access community sites to receive interventions; (5) able to take meals independently. Exclusion criteria included (1) diagnosed with cognitive impairment, indicated by a Mini Mental State Examination (MMSE) score less than 24, or (2) having dietary restrictions strictly mandated by medical prescriptions due to advanced severity of chronic conditions. Recruitment efforts involved posting flyers at community health service centers, particularly during regular physical examinations of elderly individuals.

Data Collection

The data collection for this validation study encompasses three major components. (1) Demographic information, including age, gender, economic status, healthcare coverage, and education level, were collected through a basic information questionnaire designed by the research team. (2) Health and nutritional status. general health status data, such as self-reported medical history, allergies, height, and weight, were gathered through physical measurements and basic information questionnaires; levels of physical activities and risk of malnutrition and frailty, were assessed through established

questions including IPAQ, MNA-SF, and FRAIL. (3) Dietary Behavior: two approaches were provided to collect the dietary behaviors of the community-dwelling elders. A dietary habits questionnaire extracted from CHILS questionnaire were mandatory for all participants to help understanding their dietary habits and preferences. In addition, an IoT system had been deployed in the community canteen since 2020. This system automatically tracked the elderly residents' dietary behaviors, offering a more objective and precise measure of their dietary choices.

The methods of data collection are detailed in **Table 3**, which provides a clear breakdown of the data dimensions, items, and the respective collection methods.

Table 3 Overview of demographic, dietary, health and nutrition data collection in among community-dwelling elders

Data dimensions	Data items	Collection method
Demographics	gender, age, geography, living arrangements, and monthly household income.	Basic information questionnaire designed by team
Dietary behavior ^a	Dietary diversity	IoT system log (DDS)
	Diet quality	IoT system log (CDGI-E)
Health and nutritional status	Body mass index (BMI)	Basic body measurement (body weight scale, height rod)
	Level of daily physical activities	Questionnaire (IPAC)
	Allergies, medical history	Basic information questionnaire designed by team
	Risk of malnutrition	Questionnaire (MNA-SF)
	Risk of frailty	Questionnaire (FRAIL)
Personal dietary preferences ^b	Cooking method preference, staple food preference, common oil preference, and taste preference.	questionnaire (CLHLS) [61] and/or IoT system log
	Dish preference if available	

Simulated Interventions

In the simulated intervention phase of our study, we utilized the ElCombo meal recommendation system to provide personalized dietary advice and healthy food services to community-dwelling elders (development details in previous sections). This system was driven by two key databases: a dietary behavior user profile database constructed from primary data collected from the elderly participants; an elderly food KG, constructed using historical dining behavior data from elderly residents and menus from the community canteen, with a total of 180 dish entities, 112 ingredient entities, 20 ingredient type entities, 27 nutrient entities, 30 disease entities. The main scenario considered was the lunch hour, with the daily energy intake estimated to be divided among three meals at a 3:4:3 ratio.

Participants were provided with 30 days simulated meal recommendations, offering five options each weekday. These recommendations were tailored according to the availability of their historical eating behaviors. Specifically, we categorized participants into two groups based on their recorded eating history frequency (≥ 5 times) in the IoT system. The 'tracked group', with an eating history of no less than 5 instances, received initial recommendations based on the similarity between their frequently chosen dishes and candidate dishes. The nutrient consideration were also included. For those with less than 5 eating history records, the 'untracked group', recommendations were generated by comparing their profiles to those of similar individuals with more than 5 recorded eating behaviors. This approach ensured personalized and relevant suggestions for each group. We followed the recommending principles designed in "Personalized Meal Recommendation Algorithms" section, assuming that the participants randomly took one of the recommended meals generated by ElCombo system.

Outcome Measuring

In the outcome measurement phase, we focused on evaluating the changes of diet diversity and diet quality of the community-dwelling elders over a one-month intervention period, to determine the effectiveness of the ElCombo meal recommendation system. As a simulated intervention, all data regarding participants' food selections were generated and stored locally. We retrieved this data from our repository to score the dietary scales for our analysis. For the comparison analysis, both groups set comparators using data from the tracked group. However, we compared the tracked group's recommendations to their historical eating patterns over time. For the untracked group, comparator metrics were calculated based on the tracked group's autonomous choices in which we assumed the tracked/untracked groups are with similar overall average and distributions. In addition, case analysis were conducted to examine the system's ability to tailor recommendations to the individual characteristics of community-dwelling elders.

For dietary diversity, we used the Dietary Diversity Score (DDS) [62], a measure that reflects the variety of food categories in the participants' diets ranging from 0 to 9. The non-repetitive counts of food categories contained in the combos are accumulated as the DDS. Previous studies have proved that DDS is a concise and straightforward dietary diversity indicator for older populations [53, 62-64]. The higher the DDS and/or FVS, the higher the dietary diversity and/or food diversity the lunch combos will have.

For diet quality assessment, the China Elderly Dietary Guidelines Index (CDGI-E) was used [52], with a range from 0 to 110. CDGI-E was developed based on Chinese Dietary guidelines for seniors and Food Guide Pagoda released in 2016 [45], serving as a diet quality index in large population-based elderly diet surveys [52, 53]. CDGI-E can be used to examine whether the experts experience and knowledge have been well incorporated into the recommendation system. The scoring functions of DDS and CDGI-E are provided in **Multimedia Appendix 2**.

Ethical Considerations:

This study received ethical approval from the Institute of Medical Information, Peking Union Medical College & Chinese Academy of Medical Science (Approval Number IMICAMS/01/21/HREC). Participants provided written informed consent for both primary data collection and subsequent analyses. To protect participants' privacy, de-identification was conducted by removing direct personal identifiers; essential demographic details like age and gender were securely retained for meal recommendation purposes. Personal records were stored separately, ensuring access only to authorized healthcare professionals and investigators. As a token of appreciation for their time and contribution, participants were compensated with daily necessities and health education materials valued at approximately US \$15 for the initial assessment.

Statistical Analysis

For descriptive analysis, continuous variables were presented as means with standard deviations (SDs) or median with interquartile ranges (IQRs), while categorical variables were expressed as percentages. The comparative analysis assessed diet quality and diversity changes over a 30-day intervention, using paired t-test for the tracked group and independent t-test for the untracked group. Additionally, a visualization report was generated, illustrating the outcome changes with 95% confidence intervals (CIs) for each time point. All analyses were performed using Python, version 3.8, with statistical significance set at a *P* value of less than 0.05 (two-sided).

Results

A total of 112 community-dwelling individuals were screened, among whom 96 eligible individuals were included for further data collection and analysis. Of these participants, 34 had tracked eating

history behaviors over 5 times, 62 had less than 5 times tracked eating behaviors. Two databases were incorporated into the validation procedure: an elderly-targeted food knowledge graph containing 180 dishes (see details in Methods), and a dietary -related user profiles contributed by 96elderly participants.

Overview of study sample and involved datasets

Baseline characteristics of the elderly participants by their demographics and general health status are shown in **Table 4**. More comprehensive information regarding their socio-economic status, medical history, and dietary preferences are available in **Multimedia Appendix 3 - Table 1**. The total 96 participants aged between 61 and 90 years, with an average age of 70.93 years, 45 (46.88%) being female. Particularly, we provided a deeper investigation on the multiple chronic conditions (MCCs) of our elderly participants. **Figure 6** illustrates the distribution of diagnosed diseases among the participants. Notebaly, over 60% participants suffered from 2 or more chronic conditions simultaneously, and 37.5% had 3 or more coexisting chronic conditions. The three most prevalent diseases were hypertension (59.38%), arthritis (31.25%), and diabetes mellitus (23.96%).

Table 4. Characteristics of the community-dwelling elderly participant (N=96)

Characteristics	Item	Count	Percentage (%)
Gender	Male	51	53.12
	Female	45	46.88
BMI	≤19.9	2	2.08
	20.0-26.9	66	68.75
	≥27.0	27	28.12
Age	60-64	14	14.58
	65-79	66	68.75
	≥80	16	16.67
Diagnosis of chronic conditionss	≥3	36	37.50
	2	24	25.00
	1	31	32.29
	0	5	5.21

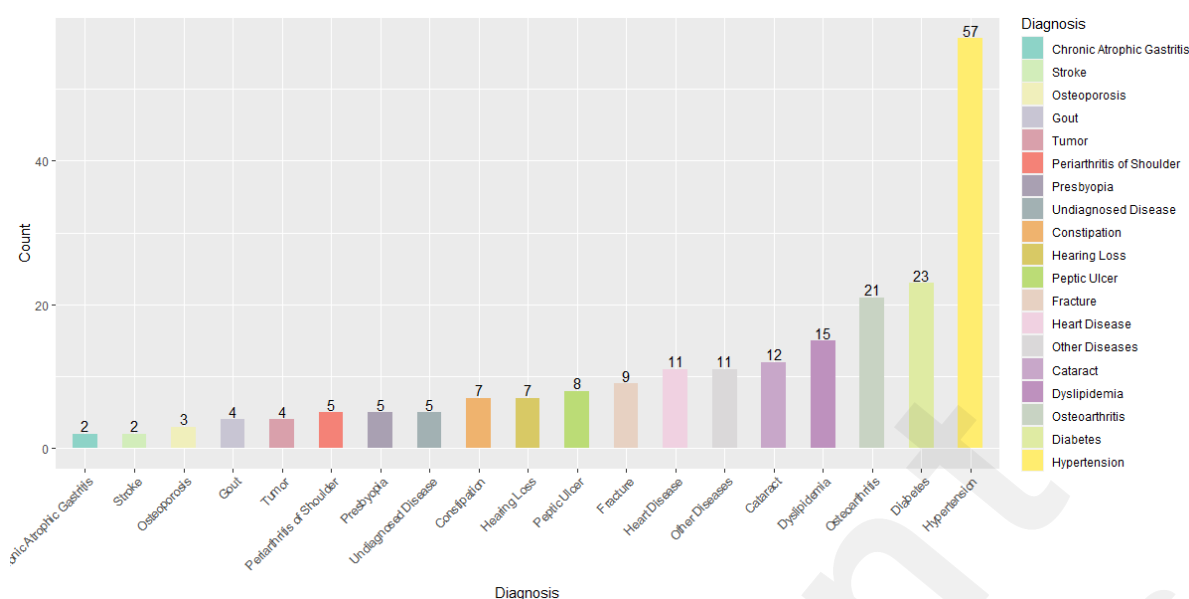


Figure 6. Distribution of comorbidity diagnoses among community-dwelling elderly participants (N=96)

For elderly-targeted food knowledge graph, **Table 5** in **Multimedia Appendix 1** presents all contained entities of 180 dishes, 112 ingredients, 20 ingredient categories, 27 nutrients, and 30 disease, which supported the personalized meal recommendations. Details regarding dish category, flavor, cooking methods, contained ingredient distribution are provided in **Figure 5** of **Multimedia Appendix 1**.

Comparative Analysis

We divided the 96 elderly participants into tracked group (N=34) and untracked group (N=62), according to whether they have over 5 times of eating history behaviors. Following this, a dual-step comparison analysis were conducted to examine the effectiveness of the simulated meal recommendation intervention. First, a pairwise and independent t-test analysis was respectively conducted to compare the DDS and CDGI-E scores before and after intervention in both groups. As shown in **Table 5**, for both groups, CDGI-E and DDS scores were reported to be statistically significantly higher in the recommended meals compared to autonomous selections ($P < .001$).

Table 5. Diet quality and diversity score improvement with meal recommendations vs. autonomous selections

Group	Indicators	Mean $\pm$ SD		t (df)	P
		Autonomous	Recommended		
Tracked	(eating CDGI-E	25.66 $\pm$ 0061	29.51 $\pm$ 0.68	24.54 (33)	<.001

history ≥ 5 times)	DDS	3.73 $\pm$ 0.12	4.9 $\pm$ 0.16	28.3 (33)	<.001
Untracked (eating history < 5 times)	CDGI-E	25.66 $\pm$ 0.61	28.05 $\pm$ 0.51	14.45(37.96)	<.001
	DDS	3.73 $\pm$ 0.12	4.58 $\pm$ 0.61	27.72 (47.04)	<.001

To assess the changes of diet quality and diversity across two groups during the simulated intervention period, we also assessed the overall trends in CDGI-E and DDS over 4 weeks. **Figure 7** illustrates time-dependent trends in dietary quality and diversity within this period. The date range was plotted on the x-axis, while the y-axis showed the values of CDGI-E and DDS. The blue lines represent the average outcome indicators of the recommended meals, while the orange lines signify those of the autonomously selected meals. Particularly, the orange line was marked as 'average' for untracked group to represent the assumption mentioned above. Given that multiple recommendations were provided as alternatives, the 95% confidence intervals of the outcome indicators are also depicted as shaded areas.



Figure 7. Visualization of 30-day simulation outcomes showing time evolution of CDGI-E and DDS score changes

From the line charts, we identified three preliminary findings. First, the outcome indicators of recommendations (blue lines) are generally higher than those of autonomous selections, demonstrating that the meals recommended by ElCombo are superior in diet quality and diversity compared to the elders' independent choices. Second, the outcome indicators for the recommended meals display less fluctuation in both CDGI-E and DDS lines than the autonomous selections,

indicating greater stability. This suggests that our proposed food-KG recommendation system can help individuals maintain a consistently high-quality and diverse diet, which might otherwise be challenging to achieve independently. Lastly, the blue lines show an increasing trend in CDGI-E and DDS scores, underlining ElCombo's potential to automatically optimize the recommended outputs over time.

Representative Case Study

We also conducted two case studies to dissect the ElCombo workflow aimed at elderly individuals. This allowed us to understand the potential benefits of a food-KG-based meal recommendation system more deeply. We randomly selected two community-dwelling elderly individuals from the previous cohort for our study – one with tracked eating history and the other without. **Figure 8** displays the health profiles and nutritional needs associated with the MCCs of the two case study subjects. We also provide final week's meal recommendations in **Multimedia Appendix 4**, which include specific dishes and their respective diet quality (CDGI-E) and diversity (DDS) scores. Additionally, for the individual with a recorded eating history, we provide a comparison between recommended and the autonomous meal choices to enhance the intuitive understanding of the personalized recommendations' impacts.

Case 1, female, age: 70-80, BMI> 28.0, Hypertension and Diabetes

Hypertension	• Salt(<5g/d) • Cholesterol(<300mg/d) • Fats(<30g/d)
Diabetes Overweight	• Caloric intake(20-25kcal/kg) • Percentage of energy intake at lunch(30-35%) • Percentage of energy provided by fats(<25) • Prioritize low glycemic index (GI) carbohydrate sources

Case 2, male, age: 60-70, BMI: 25.0-27.0, Hypertension, Hyperlipidemia, Hypercholesterolemia, and Diabetes

Hypertension Hypercholesterolemia	• Salt(<5g/d) • Cholesterol(<200mg/d) • Fats(<30g/d)
Diabetes Normal Weight	• Caloric intake(25-30kcal/kg) • Percentage of energy intake at lunch(30-35%) • Percentage of energy provided by fats(<25) • Prioritize low glycemic index (GI) carbohydrate sources

Figure 8. Health profiles and nutritional needs linked to the MCCs of case study subjects

Our first case study focused on a female participant in her seventies with a BMI over 28.0, diagnosed with hypertension and diabetes. Her user profile data included a preference for salty flavors, a staple diet based on rice, and light daily physical activity. Adhering to the designed framework, we began

by filtering out candidate meals that failed to meet elderly nutrition guidelines, particularly those related to her health conditions. Specifically, we imposed limits on salt, cholesterol, and oil content, dictated by “restrict” relations for these food entities. Her diabetes diagnosis and daily physical activity level also dictated energy intake (20-25 kcal/kg), energy proportion from lunch (30-35%), and energy proportion from fat (<25%). We generated the top five meal candidates per weekday based on nutrient adequacy ranking. These candidates adhered to both disease-related nutrition requirements and elderly nutrition guidelines.

Upon comparing the recommended meals with those the subject selected independently over a week (5 weekdays), we noticed the increased diversity of dishes in the recommended meals. The recommended combos contained 17 dishes, compared to 9 dishes in the combos she selected. Our recommendations also provided more variety in staples and soups. This led to a marked increase in diet diversity and quality. Furthermore, the recommended meals had a high proportion of lower glycemic index carbohydrates such as whole grains, reflecting the tailored recommendation rules for her health conditions.

Our second case study examined a male participant in his sixties with a BMI between 25.0 and 27.0, diagnosed with hypertension, diabetes, and dyslipidemia. He preferred lighter flavors and rice-based meals. As his eating behavior history was unavailable, we used a reference elderly individual (case 1) with the highest Jaccard similarity of 0.7. Recommendations for the male subject were further refined based on the reference individual's most frequently selected dishes. Health-related restrictions about diabetes, hypertension, and dyslipidemia helped us set limits on cholesterol (<200mg/d) and energy intake (20-25kcal/kg). We excluded those meals that did not meet disease-related nutrition requirements, resulting in 282 suitable meals out of the initial 708 candidates.

Our system generated 17 unique dishes in various combinations for a 5-day meal plan, with a repetition rate of only 25%. Few meat-based dishes were recommended to the male participant than the female, possibly due to the cholesterol-related preferences identified in his user profile. Despite variations in the recommended dishes, both meal sets demonstrated similar diet diversity and quality. This suggests that ElCombo can effectively generate diverse, high-quality meal recommendations for community-dwelling elderly individuals. This is achievable with minimal reliance on eating history while maintaining a keen focus on personal health status.

Discussion

Principal findings

Healthy eating is crucial in mitigating MCCs and promoting healthy aging [1, 65]. However, finding suitable solutions that cater to the specific needs of community-dwelling elders remains a challenge [4]. In this study, we developed ElCombo, a personalized meal recommendation system with specific attention to the unique challenges community-dwelling elders face. We demonstrated significant promise of ElCombo in improving the dietary behavior of community-dwelling elders, particularly in terms of diet diversity and quality. This system offers a highly tailored and adaptable solution to the dietary challenges faced by this demographic. ElCombo represents a responsive approach to the evolving dietary habits of the elderly users, ensuring that recommendations remain relevant and effective.

The development of our system involved three key steps: constructing a food-KG for nutritional and disease-related knowledge support, establishing user profiles to match individual needs with food intakes, and developing a recommendation algorithm that allows updates based on dietary behavior changes. The use of automated technologies in our system makes it a cost-effective approach to individual dietary management [66]. Our experiments reveal that ElCombo outperforms autonomous dietary choices, with elders showing improved performances in diet diversity and quality, as measured by DDS and CDGI-E. These findings underscore ElCombo's effectiveness as a tool for promoting healthier dietary habits and enhancing health outcomes among the community-dwelling elders. Its time awareness allows the recommendations to adapt with the dietary needs of community-dwelling elders with MCCs.

Strengths and implications

Compared to other food recommendation systems, ElCombo has unique strengths and innovations, stemming from its focus on the daily dietary needs of community-dwelling elderly individuals. Our research focus and designated approach is backed by distinctive data sources. Specifically, we collected data on the eating choices of community-dwelling elders from menus in community canteens. Additionally, we gathered primary data on their health status and dietary preferences to ensure a more rigorous and reliable validation study. This approach effectively balances the need for nutrient intake that meets disease-related requirements with the flexibility to cater to personal preferences, offering a more sustainable impact on elders' dietary health. Unlike previous systems that primarily offered guidance on specific food groups or nutrients [67], ElCombo provides more comprehensive and practical meal-based advice for daily eating scenario[42, 68]. This highlights a key feature of ElCombo - user-friendliness, which displays a variety of well-organized meal which

integrates different dishes instead of a lengthy list of individual food items.

Although some existing food recommendation systems have previously offered meal-based suggestions[26, 28, 31], ElCombo stands out with a more comprehensive and tailored considerations of elders. It prioritizes the complex health conditions common among the elderly, particularly those with MCCs. This has led to the development of a rigorous knowledge base and recommendation algorithm, enabling the system to address the broad spectrum of nutritional needs. Additionally, ElCombo excels for its attention to food culture. Unlike systems that focus solely on main ingredients[31, 69], ElCombo includes a detailed analysis of side ingredients and seasonings, ensuring a more comprehensive nutrient composition and precise recommendations. Finally, ElCombo demonstrates high compatibility with diverse socioeconomic statuses by offering diverse combinations with different dish types and quantities to suit varying food budgets.

The elderly targeted food KG built in this study, combining expert knowledge with user profiles and reasoning rules, enables ElCombo to handle individual case queries effectively. Similar to the findings from previous work, knowledge-based food recommendation approach proves reliable in this study. The reliance on a rule-based recommendation pipeline in ElCombo is justified by its ability to manage complex interactions between food constraints and nutritional requirements, providing stable and streamlined meal recommendations.

Transforming complex clinical guidelines into detailed initiatives in community settings and home environments enhances the effectiveness of elderly care informed by abstract expert knowledge, leading to a profound impact on public health [55, 70]. Accordingly, the potential benefits our proposed system, which is tailored for daily meal recommendations for community-dwelling elders, can be both durable and adaptable. Its methodology can be extended to other self-management practices in these community-dwelling elders, including medication adherence, physical activity, and mental wellness programs. This personalized recommendation approach, characterized by comprehensive knowledge support and sensitive adaptation to individual needs, is especially beneficial for elderly individuals living with MCCs, helping them achieve a higher quality of life in their aging years.

Limitations and Future Work

Despite its promise, the system does have limitations and areas for future development. The evaluation experiment was conducted under a simulation setting with a retrospective study design;

also, the present system was designed to provide advice on recommended intake rather than actual intake [23, 71]. These two concerns raise questions about the real-world effectiveness of our system. Future work will address this by conducting real-world clinical trials to evaluate the system's effectiveness, acceptability, and user experience [72]. Additionally, we recognize that the current data extraction efficiency from large corpora is a challenge, especially considering the complex health conditions and prescriptions often associated with older adults. In response, we plan to enhance our methodology by implementing deep neural networks for named entity recognition (NER) and relation extraction (RE), thereby increasing the expansion speed of food-KG [33]. This improvement will provide a more nuanced approach to addressing the unique health and prescription needs of the elderly population.

Conclusion

In this study, we developed and evaluated ElCombo, a personalized meal recommendation system designed for Chinese community-dwelling elders. Our findings in simulated environments revealed that ElCombo hold the promise to effectively improve the dietary diversity and quality of the community-dwelling elders, as reflected by the increase of DDS and CDGI-E scores. While these results are promising, they are preliminary and situated within a controlled, simulated context. The true effectiveness of ElCombo to enhance dietary practices and potentially impact health outcomes among the aging population remains to be established through rigorous real-world clinical trials. Consequently, future research is dedicated to conducting more comprehensive feasibility and effectiveness evaluations among community-dwelling elders to confirm the system's efficacy, and to refine its ability to process complex health data effectively. The adaptability and preliminary cost-effectiveness observed in this study suggest that ElCombo has the potential to be a valuable tool in improving the dietary health of elderly individuals, pending further validation.

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Authors' contribution

JL conducted the study. ZX and XX designed the study framework. ZX developed the system and drafted the manuscript. All other authors contributed to the study conceptualization, data collection and experiment results interpretation. XX, ZG, HK and LS contributed to the data collection and result analysis. YG contribute to the food and user profile data cleansing. ZX and YG contributed to the simulation experiments. MT provided insightful suggestions for draft revision. All authors revised the manuscript for important intellectual content and approved the final manuscript.

Competing interest

All authors declare no financial or non-financial competing interests.

Data availability

The datasets used and/or analyzed during not publicly available for the data privacy concern, but maybe available from the corresponding author on reasonable request.

Disclosure of AI Use

The authors attest that there was no use of generative artificial intelligence (AI) technology in the generation of text, figures, or other informational content of this manuscript.

Multimedia Appendix 1

Food-KG Data Preparation

Multimedia Appendix 2

The Personalized Meal Recommendation Algorithm and Outcome Scoring Functions

Multimedia Appendix 3

Characteristics of Community-dwelling Elderly Participants

Multimedia Appendix 4

Instances of Personalized Meal Recommendations

Abbreviations:

MCC: multiple chronic conditions

Food-KG: food knowledge graph

CLHLS: Chinese Longitudinal Healthy Longevity Survey

CDGI-E: China Elderly Dietary Guidelines Index

DDS: diet diversity score



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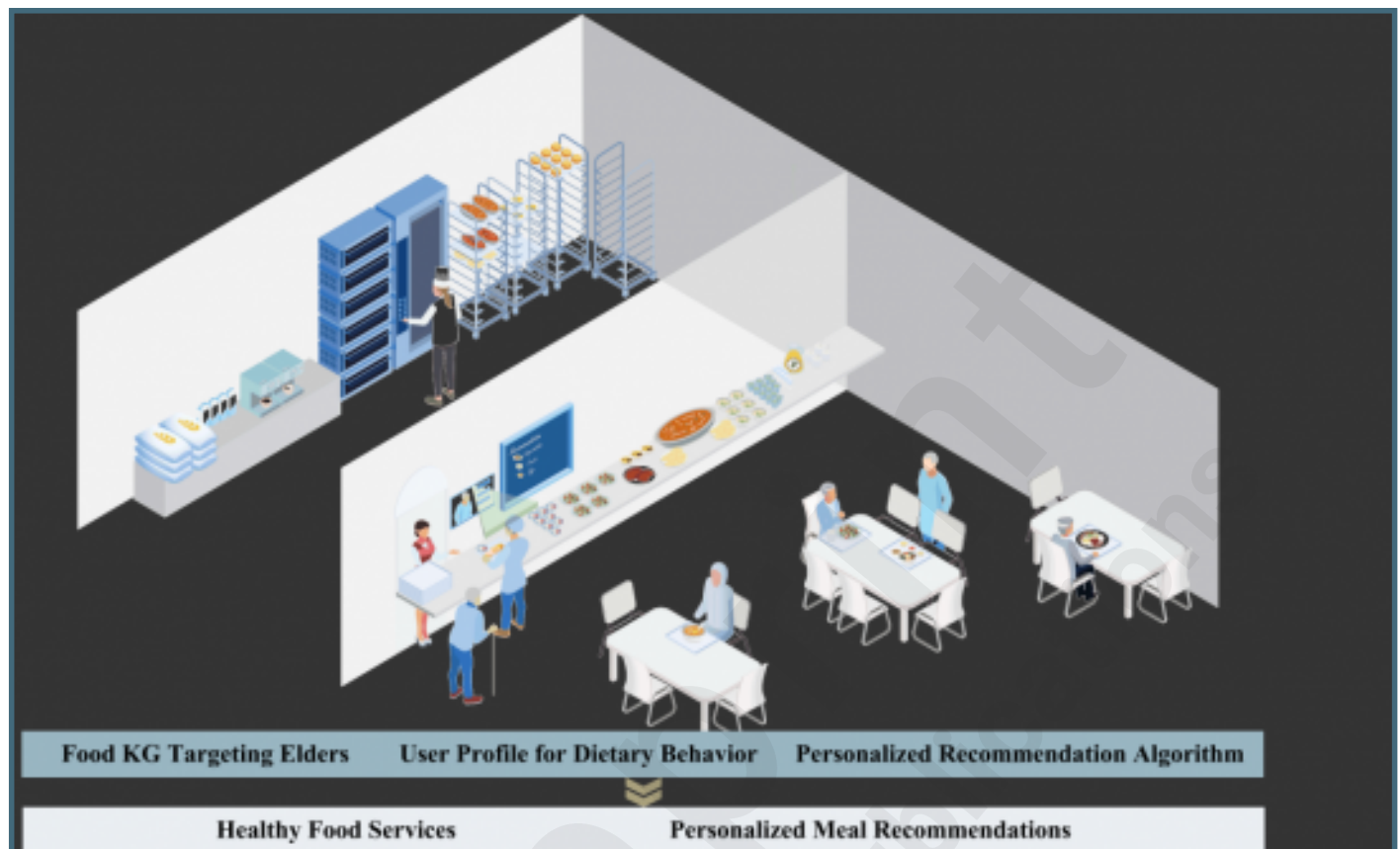
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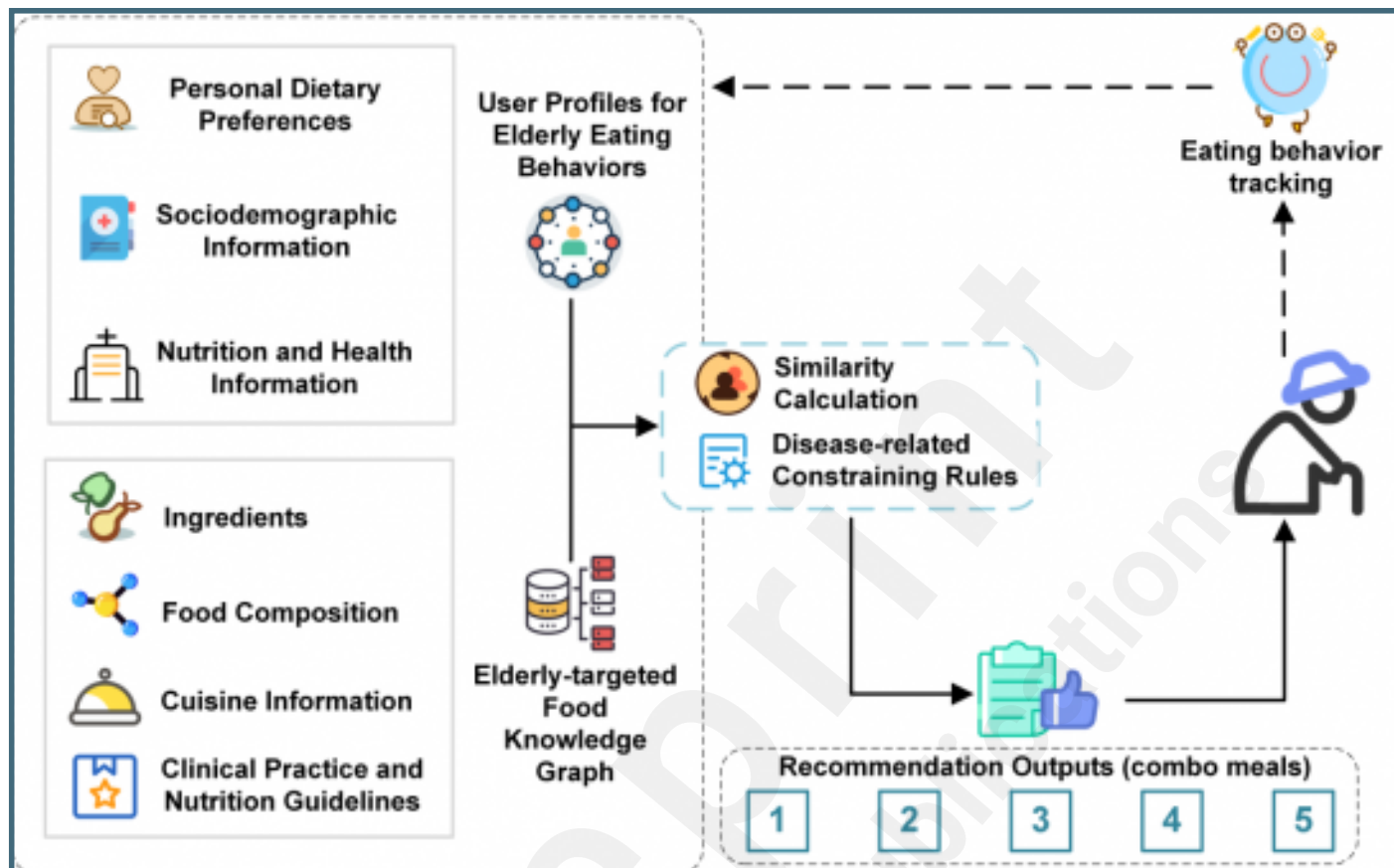
Supplementary Files

Figures

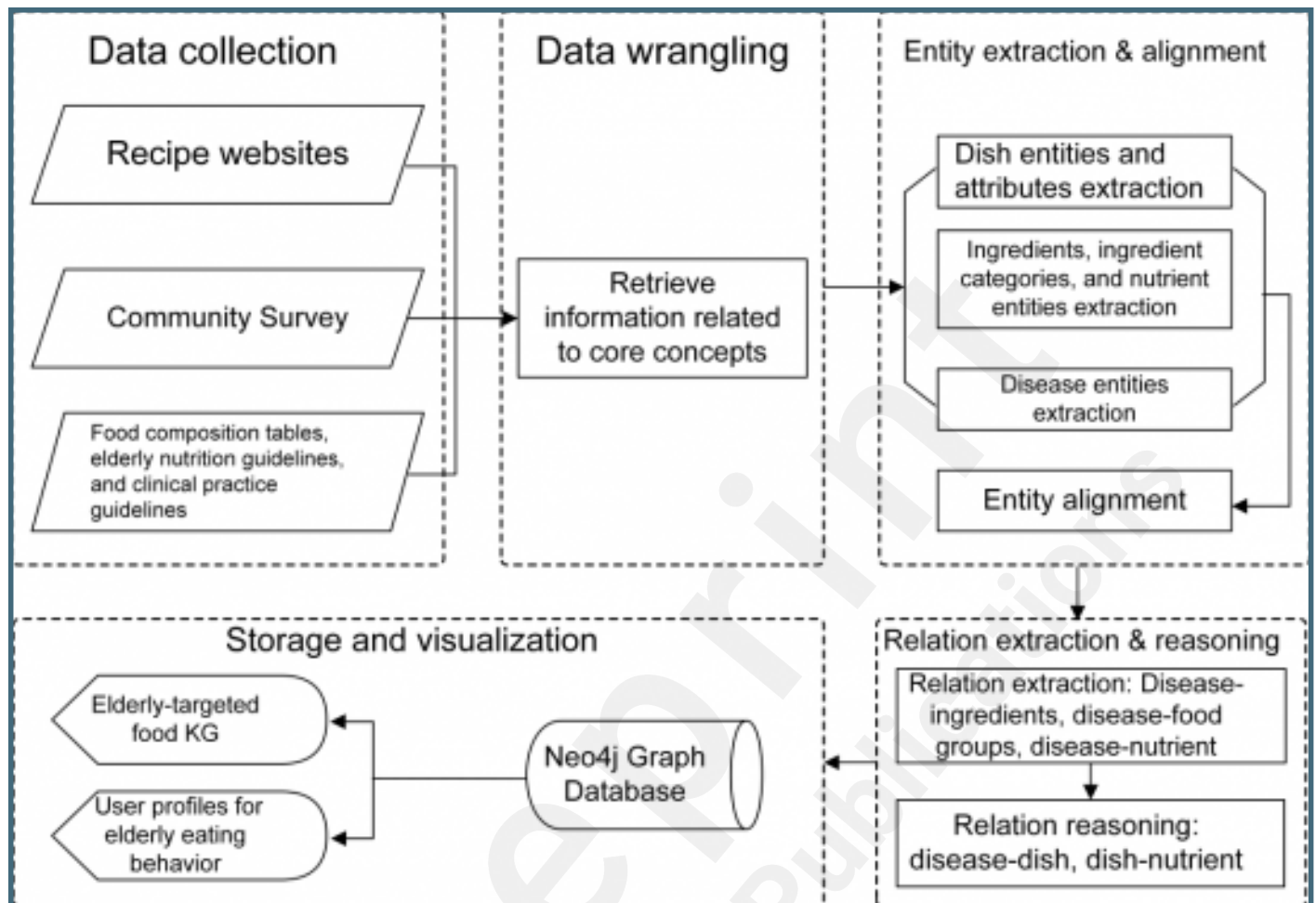
A scenario landscape of ElCombo application in personalized meal recommendation community-dwelling elderly care.



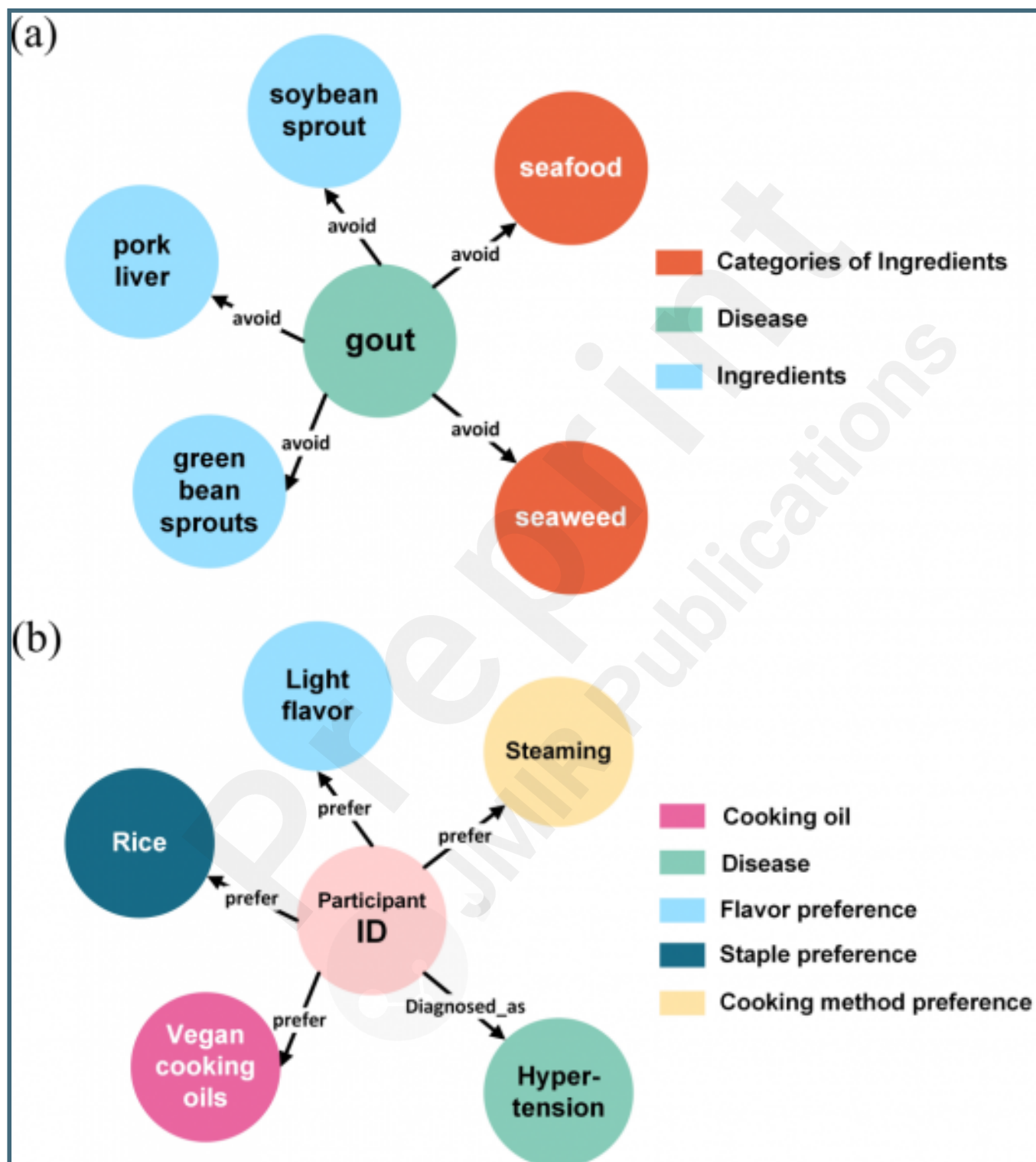
Overview of ElCombo's architecture for elder-specific food recommendation, including knowledge graph and algorithm modules.



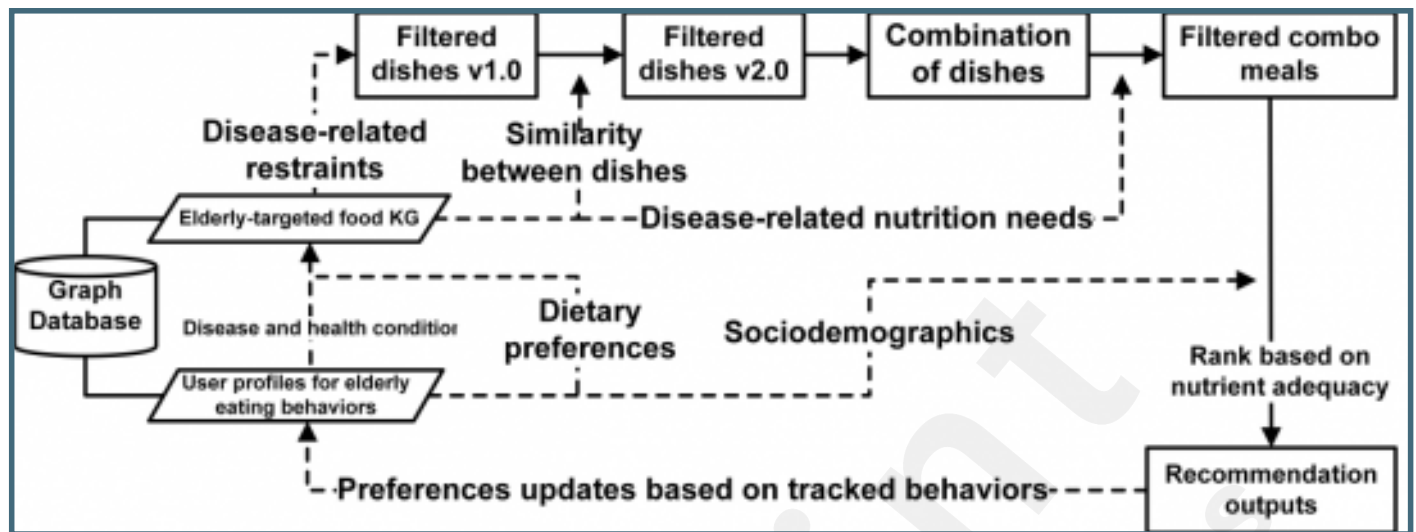
Food knowledge graph construction pipeline.



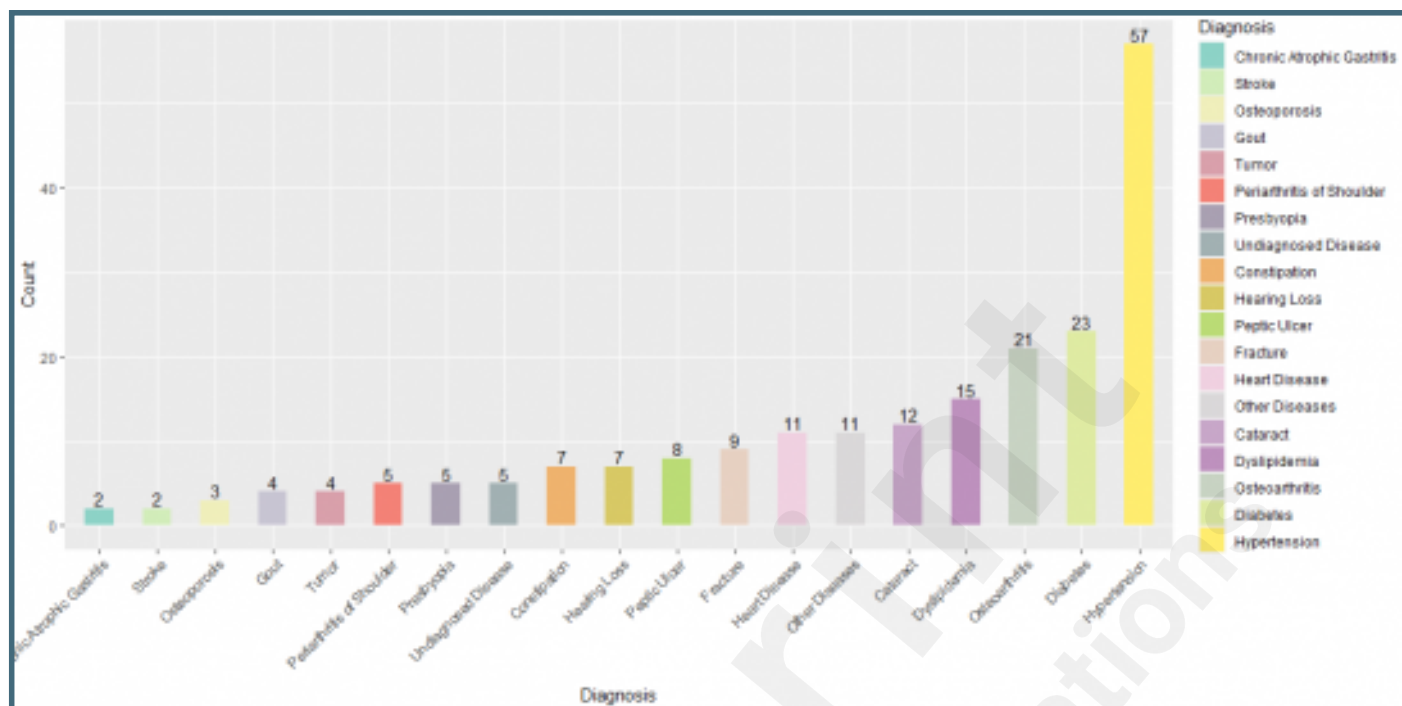
Visualizing disease-diet connections and individual health profiles in Food-KG. (a). Mapping 'gout' disease to relevant dietary ingredients and categories; (b). Elderly profile with hypertension, dietary preferences interconnections depicted.



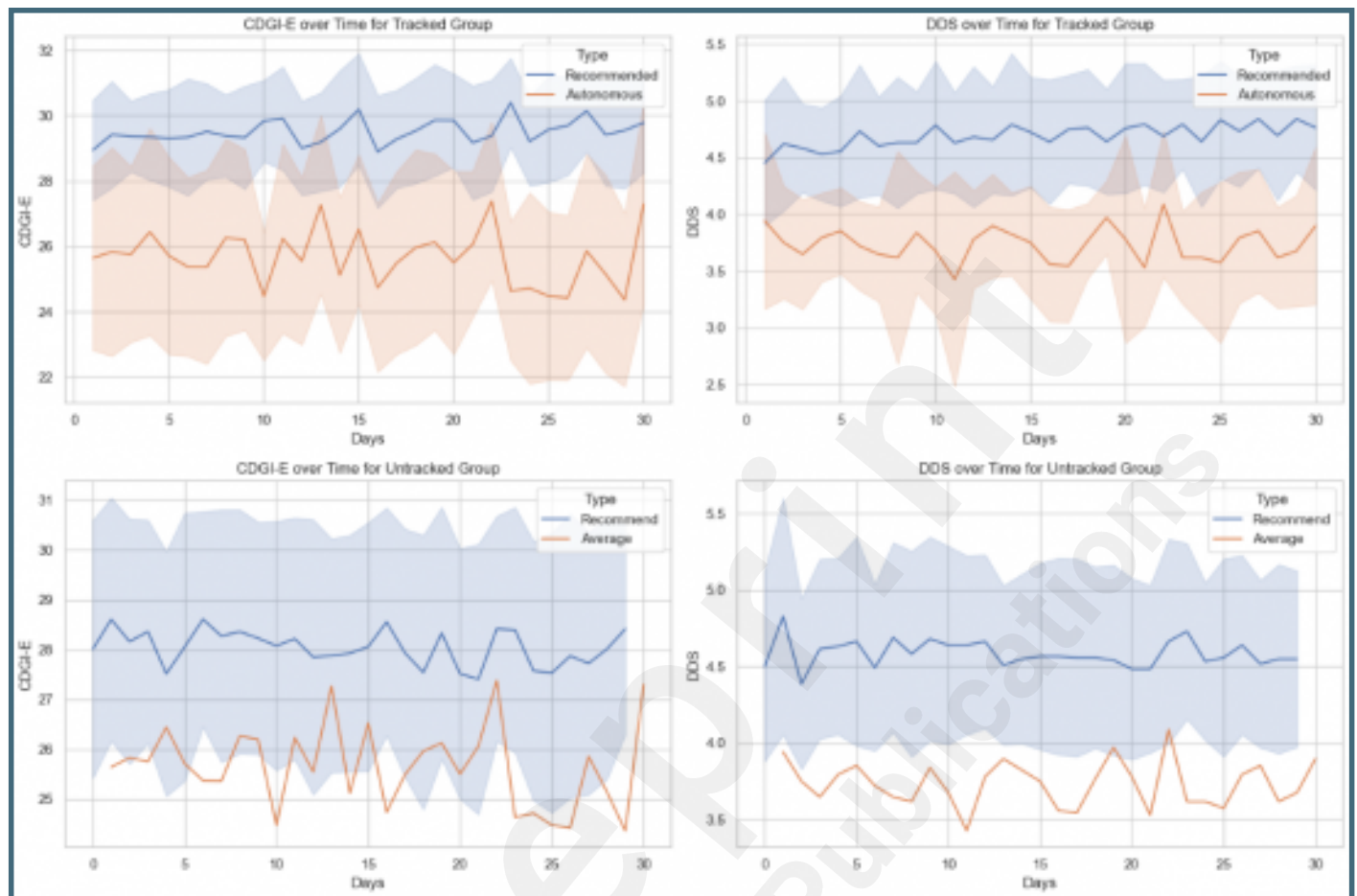
Dual-stage personalized meal algorithm: dish generation and combo meal recommendations.



Distribution of comorbidity diagnoses among community-dwelling elderly participants (N=96).



Visualization of 30-day simulation outcomes showing time evolution of CDGI-E and DDS score changes.



Health profiles and nutritional needs linked to the MCCs of case study subjects.

Case 1, female, age: 70-80, BMI> 28.0, Hypertension and Diabetes	
Hypertension	• Salt(<5g/d) • Cholesterol(<300mg/d) • Fats(<30g/d)
Diabetes Overweight	• Caloric intake(20-25kcal/kg) • Percentage of energy intake at lunch(30-35%) • Percentage of energy provided by fats(<25) • Prioritize low glycemic index (GI) carbohydrate sources
Case 2, male, age: 60-70, BMI: 25.0-27.0, Hypertension, Hyperlipidemia, Hypercholesterolemia, and Diabetes	
Hypertension Hypercholesterolemia	• Salt(<5g/d) • Cholesterol(<200mg/d) • Fats(<30g/d)
Diabetes Normal Weight	• Caloric intake(25-30kcal/kg) • Percentage of energy intake at lunch(30-35%) • Percentage of energy provided by fats(<25) • Prioritize low glycemic index (GI) carbohydrate sources

Multimedia Appendixes

Food-KG Data Preparation.

URL: <http://asset.jmir.pub/assets/6c96d5bcc2f86c1d2769231a86fa89b0.pdf>

The Personalized Meal Recommendation Algorithm and Outcome Scoring Functions.

URL: <http://asset.jmir.pub/assets/95c2800b2ca4b9e54f2a860be73e99e2.pdf>

Characteristics of Community-dwelling Elderly Participants.

URL: <http://asset.jmir.pub/assets/8399217f486f68b3773f25adbb76db8f.pdf>

Instances of Personalized Meal Recommendations.

URL: <http://asset.jmir.pub/assets/75f78f6a6e2a6fb1975e99a2b4b48baf.pdf>



Related publication(s) - for reviewers eyes onlies

This is a compressed file containing ethical consideration documents for the review process, not intended for publication. It includes 1) the original ethical approval and its translation, and 2) the informed consent form, adapted to the literacy and reading levels of the elders, along with its translation.

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